



* Practical - 6 *

* activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Progress" />

    <ProgressBar
        android:id="@+id/progressBar"
        style="@android:style/Widget.ProgressBar.Horizontal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:max="100"
        android:progress="0"
        android:layout_marginTop="20dp" />

    <TextView
        android:id="@+id/txtProgress"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Progress: 0%"
        android:layout_marginTop="10dp" />

    <TextView
        android:id="@+id/txtMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text=""
        android:textStyle="bold"
        android:layout_marginTop="20dp" />
</LinearLayout>
```



* MainActivity - java

```
package com.example.program6;
```

```
import android.os.AsyncTask;
```

```
import android.os.Bundle;
```

```
import android.widget.Button;
```

```
import android.widget.ProgressBar;
```

```
import android.widget.TextView;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    Button btnStart;
```

```
    ProgressBar progressBar;
```

```
    TextView txtProgress, txtMessage;
```

⑨ Override

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        btnStart = findViewById(R.id.btnStart);
```

```
        progressBar = findViewById(R.id.progressBar);
```

```
        txtProgress = findViewById(R.id.txtProgress);
```

```
        txtMessage = findViewById(R.id.txtMessage);
```

```
        btnStart.setOnClickListener (v -> {
```

```
            new ProgressTask().execute();
```

```
        });
```

```
}
```



```
private class ProgressTask extends AsyncTask<Void,  
Integer, String> {
```

① Override

```
protected void onPreExecute() {  
    super.onPreExecute();  
    progressBar.setProgress(0);  
    txtProgress.setText("Progress: 0%");  
    txtMessage.setText("");  
}
```

② Override

```
protected String doInBackground(Void... voids) {  
    int progress = 0;  
    while (progress <= 100) {  
        try {  
            Thread.sleep(500);  
        } catch (InterruptedException e) {  
            e.printStackTrace();  
        }  
        publishProgress(progress);  
        progress++;  
    }  
    return "☑ Task completed Successfully!";  
}
```



⑨ Override

```
protected void onProgressUpdate (Integer  
values) {
```

```
    int value = values[0];
```

```
    progressBar.setProgress(value);
```

```
    txtProgress.setText("Progress:" + value  
                        + "%");
```

```
}
```

⑩ Override

```
protected void onPostExecute (String result) {
```

```
    txtMessage.setText(result);
```

```
}
```

```
}
```

```
}
```